Springs, Simple Pendulums, and Damped Oscillations — Examples of Oscillatory Motion

SYNOPSIS

We explore a few examples of oscillatory motions: Springs, and simple pendulums.

BEFORE YOU READ ON — PREDICT

A pendulum clock keeps time by swinging a bob on a string. Three things you could change: **use a heavier bob**, **use a longer string**, **take it up a mountain**. Which of those change how long a swing takes? Write down your three answers before you read on. One of them catches almost everybody.

PERFORMANCE EXPECTATIONS

1. **Solve** the spring equation of motion and obtain $\omega = \sqrt{(k/m)}$.
2. **Show** that a pendulum at small angles is simple harmonic, with $\omega = \sqrt{(g/L)}$.
3. **State** the small-angle approximation and say at what angle it stops being safe.
4. **Explain** why a pendulum's period does not depend on the mass of the bob where a spring's does.
5. **Describe** what damping does to the amplitude of an oscillation, and what it very nearly does not do to the period.

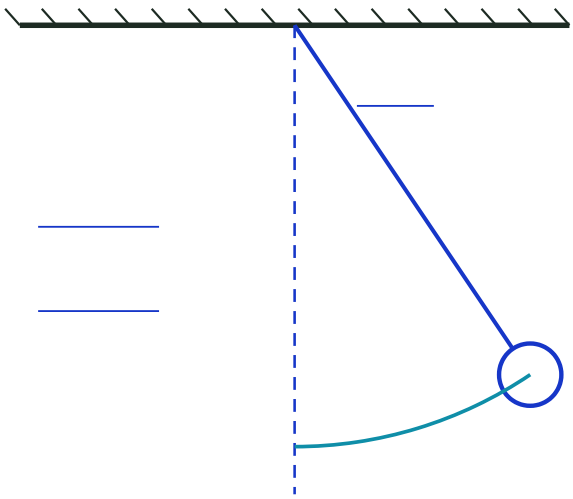
Spring Oscillation

Note 4 got as far as $ma = -kx$: the force is proportional to the displacement. That is a statement about the system, and it is not yet a statement about *time* — nothing in it says how long a swing takes. Getting the period out means treating the acceleration as what it is, the second derivative of position, and then **solving** the result rather than describing it.

1. Write the acceleration as a derivative of position, in two steps, and so turn $a = -(k/m)x$ into an equation of motion. What kind of equation is it?
2. Guess a solution of the form $x(t) = A\cos(\omega t + \varphi_0)$, where A, ω and φ_0 are constants. Differentiate it once to get v, and again to get a.
3. Put your expression for a back into the equation of motion and cancel what you can. What does ω have to be?
4. Turn ω into a frequency. What happens to f if the spring is stiffer? If the mass is larger?

Simple Pendulum

The same question for a different restoring force. A bob on a string of length L, pulled aside through an angle θ: there is no spring here, so what pulls it back? Gravity does — but only the part of it that lies *along the arc*. The string takes care of the rest.



Resolve the weight on this figure. Which component moves the bob along the arc, and what balances the other one?

5. Write Newton's second law for the tangential direction only. Which forces appear, and which do not?

6. Resolve the weight and put it in. Then cancel the mass. What is a_t?

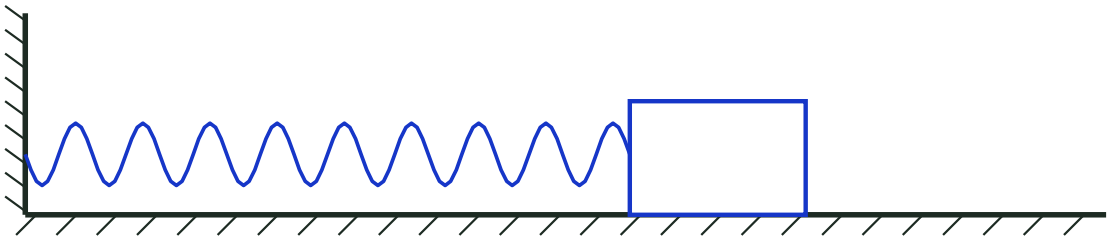
7. For a small swing, replace sin θ with θ, and then replace θ using the arc length. What equation of motion do you get?
8. Compare it with the spring's equation of motion. Write down ω without doing any more work.

THE PUNCHLINE

Two completely different systems, one equation of motion, one solution. **Neither frequency contains the amplitude**, and the pendulum's does not contain the **mass**. A heavy bob and a light bob on the same string keep the same time, which is exactly why a pendulum makes a usable clock.

Damped Oscillation

Real oscillations stop. Add one more force — a drag that opposes the motion and grows with speed — and the same method still works, with one more term to carry.



A block sliding on a surface, with the spring pulling it back and drag opposing the velocity. Drag depends on **how fast** it is going, not on where it is: $F_{\text{drag}} = -bv$.

9. Write the net force with both the spring and the drag in it, and set it equal to ma.

10. Turn it into an equation of motion by writing both a and v as derivatives of x, and gather everything on one side.

11. The solution is $x(t) = Ae^{-(b/2m)t}\cos(\omega t + \varphi_0)$. Which part of that is the oscillation and which part is the dying away?

12. What does a large $b/2m$ do? A small one? And what stays the same as the amplitude falls?



Sketch the motion here. Mark the envelope and show where you would measure the period.

Looking ahead

Three systems, one method: write the net force, turn it into an equation of motion, solve it. You now have a frequency for a spring and for a pendulum, and a description of what happens when friction is let in. **Note 6** turns all of this on an actual problem — a block on a spring with real numbers, where you have to work backwards from a graph to find ω and then to a force. After that, **Note 7** leaves the single oscillating object behind and lets the disturbance travel.

Where this sits. Every one of these three is Newton's second law with a different force law put into it. Nothing about oscillation came from anywhere else — it is all in the minus sign, which is what sends the system back toward where it started.

BACK TO YOUR PREDICTION

A heavier bob: no change. The mass cancels in step 6, so it is not in $\omega = \sqrt{(g/L)}$ at all. **A longer string: slower.** L is in the denominator. **Up a mountain: slower.** g is a little smaller up there and g is on top, so ω falls, the swings take longer, and a pendulum clock carried up a mountain loses time. The heavy-bob one is what catches people — the same instinct that says heavy things fall faster.